

# Autonomous Agentic Trading Production Blueprint - 2026 (Expanded Research)

This is the production-grade expanded research for building true autonomous (not simple bot) agentic trading systems for stocks, crypto, FX, futures, options, and multi-asset portfolios from scratch to production.

Key expansions beyond initial report:

- Full layered architecture: perception, specialist analysts, structured debate (states like InvestDebate/RiskDebate), decision/execution, guardian safety, reflection/meta-learning, memory tiers (episodic + vector + procedural + long-term), temporal orchestration.
- Multi-market data & execution: Unified models, real-time + fundamentals feeds, brokers (Alpaca with official 2026 MCP Server for natural-language agent trading across stocks/options/crypto is highlighted as ideal; IBKR, CCXT), realistic impact/slippage/latency modeling.
- Realistic simulation (critical): QuantReplay (open-source multi-asset LOB matching engine + synthetic order flow + latency + slippage for equities/FX/futures/digital), ABIDES-MARL, PyMarketSim, FinRL-Meta. Historical backtests alone are insufficient; use LOB replay and synthetic flow for production confidence.
- Safety for autonomy: Deterministic guardrails and kill switches MUST be outside the LLM (pre-action checks, step limits, global+per-strategy mutex, circuit breakers). Execution layer is binary; intelligence is probabilistic.
- Production ops: Docker, OTEL + rich COT, checkpoint resume, cost control (selective LLM), monitoring, staged rollout with paper+sim gates.
- Phased roadmap: 0 (foundation data/rules/paper), 1 (agentic core debate/memory), 2 (realism sim + guardrails + meta), 3 (live broker/MCP + observability + regulatory package), 4 (scaling autonomy).
- Framework insights: LangGraph excellent for stateful debate (see TradingAgents patterns), but Rust custom (tredo style or Nautilus Trader) for production safety/determinism/low-resource. tredo already has strong alignment (hierarchy, episodes + regret + meta, DisciplinedCore rules with memory, temporal, skills, COT).
- Risks & lessons: Overfitting, reproducibility, realism gap, safety escapes, regulatory liability (deployers responsible), cost explosion. Mitigations: sealed tests, grounded data, proper simulators, kill switches + pre-gates, full immutable audit.

See the companion .md for full details, code patterns, tredo-specific gap analysis, and references (arXiv 2605.19337, TradingAgents full repo/code, QuantReplay, Alpaca MCP, guardrails papers, etc.).

Not financial advice. For research and safe system building only.